

HOQUIAM BOWERMAN, WA, USA

WMO: 727923

Lat: 46.973N

Lon: 123.930W

Elev: 12

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 94225

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	28.5	30.8	17.7	13.4	33.1	22.7	17.1	35.2	34.5	50.9	31.0	50.5	8.1	70	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.0	78.3	63.8	72.9	61.8	69.4	61.1	65.4	75.0	63.6	70.4	62.3	67.6	9.5	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.3	84.1	67.2	61.1	80.4	65.5	59.9	77.1	64.4	30.1	75.3	28.9	70.2	27.9	67.1	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.3	21.5	18.6	DB	24.2	88.9	3.5	4.0	21.7	91.7	19.6	94.1	17.7	96.3	15.1	99.2
			WB	22.7	69.9	3.9	2.0	19.9	71.4	17.7	72.5	15.5	73.7	12.7	75.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	43.4	43.8	46.1	48.8	53.5	57.5	60.6	61.3	59.3	52.8	46.5	42.5	(6)
	DBStd	8.03	5.10	4.80	4.31	4.10	4.34	3.73	3.39	3.16	3.92	4.18	5.18	5.10	(7)
	HDD50	986	210	176	136	68	13	0	0	0	0	20	127	236	(8)
	HDD65	5003	671	592	587	487	359	231	143	124	178	377	556	698	(9)
	CDD50	1490	4	4	14	31	121	225	330	350	278	109	21	3	(10)
	CDD65	29	0	0	0	0	2	5	8	8	6	0	0	0	(11)
	CDH74	356	0	0	0	7	37	61	82	97	71	1	0	0	(12)
	CDH80	96	0	0	0	1	9	17	26	28	15	0	0	0	(13)
(14) Wind	WSAvg	8.9	10.3	9.7	9.9	9.2	8.6	8.5	8.0	7.4	7.0	8.1	9.7	10.4	(14)
(15) Precipitation	PrecAvg	70.3	9.6	8.4	7.4	4.6	3.2	2.0	1.3	1.7	3.5	6.3	10.9	11.0	(15)
	PrecMax	83.7	14.8	19.1	12.7	9.9	7.8	4.8	4.4	4.8	10.7	14.2	28.8	16.1	(16)
	PrecMin	53.4	0.4	3.1	1.4	1.8	1.0	0.6	0.0	0.0	0.1	1.4	2.5	4.7	(17)
	PrecStd	7.7	4.3	3.8	2.8	1.8	1.6	1.2	1.3	1.3	2.2	3.3	5.0	3.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.3	59.5	65.1	73.1	79.4	81.9	83.6	83.4	82.3	71.7	59.5	55.2	(19)
		MCWB	50.9	49.1	52.8	55.8	61.1	64.3	67.4	67.1	63.4	57.2	55.6	52.8	(20)
	2%	DB	53.3	55.1	58.7	63.4	69.7	71.5	74.6	75.1	74.9	65.3	57.3	52.9	(21)
		MCWB	50.9	48.4	50.7	53.2	57.8	61.9	63.7	63.8	62.1	56.5	54.8	51.1	(22)
	5%	DB	51.9	52.6	54.8	58.5	63.4	66.3	70.0	70.3	69.8	61.7	55.7	51.2	(23)
		MCWB	50.0	48.7	49.8	51.3	56.4	58.9	62.2	62.5	61.3	56.5	53.5	49.4	(24)
	10%	DB	50.4	50.6	52.6	55.3	60.1	63.5	67.1	67.6	66.1	59.4	54.2	49.7	(25)
		MCWB	48.6	47.7	48.8	50.1	54.3	57.7	60.6	61.3	60.0	55.6	52.2	48.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.4	53.3	54.9	58.2	63.0	66.2	68.7	68.0	66.3	61.4	57.4	54.0	(27)
		MCDB	53.9	55.7	60.6	68.2	73.6	77.3	80.9	80.4	75.8	65.3	58.4	54.8	(28)
	2%	WB	51.8	51.2	52.6	54.7	59.4	62.3	64.7	65.0	63.9	59.2	55.6	51.4	(29)
		MCDB	52.3	52.9	56.4	60.6	67.3	70.5	72.6	74.0	71.0	62.3	56.7	52.1	(30)
	5%	WB	50.5	49.8	51.1	52.5	57.0	60.0	63.0	63.4	62.3	57.5	54.2	50.0	(31)
		MCDB	51.3	51.5	53.9	56.6	62.8	65.1	68.7	68.8	67.5	60.2	55.3	50.9	(32)
	10%	WB	48.9	48.1	49.7	51.1	55.1	58.4	61.4	62.1	61.1	56.2	52.6	48.1	(33)
		MCDB	50.0	50.1	52.0	54.4	59.3	62.7	66.2	66.6	65.2	58.9	53.8	49.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	10.9	11.8	12.5	12.4	11.7	12.3	13.0	14.1	12.9	10.5	8.7	(35)
		MCDBR	9.4	14.0	16.9	20.3	21.2	19.8	20.2	20.9	23.3	17.2	11.0	8.9	(36)
	5% WB	MCWBR	7.3	9.2	10.4	12.0	12.3	11.1	11.4	11.4	12.6	10.5	8.3	7.6	(37)
		MCWBR	7.3	9.7	13.5	15.9	17.6	16.9	17.2	17.6	17.2	12.7	8.3	7.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.331	0.324	0.340	0.373	0.407	0.378	0.358	0.362	0.349	0.341	0.335	0.325	(40)
		taud	2.465	2.527	2.495	2.369	2.278	2.398	2.486	2.474	2.510	2.525	2.469	2.457	(41)
	Ebn at Noon		239	270	281	280	272	281	286	280	274	259	235	227	(42)
		Edh at Noon	22	24	29	36	41	36	33	32	28	24	21	19	(43)
(44) All-Sky Solar Radiation	RadAvg		333	608	921	1345	1658	1811	1923	1654	1269	739	397	279	(44)
	RadStd		48	77	103	120	141	160	151	127	109	81	50	37	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page